^{27}Al(^{14}N,αpnγ) 1976Wa11

1976Wa11: E=40 MeV ^{14}N beam was produced at the Brookhaven National Laboratory. Target was ^{27}Al. γ rays were detected with Ge(Li) detectors. Measured Eγ, Iγ, $\gamma(\theta)$, $\gamma\gamma$-coin, recoils distances. Deduced levels, J, π, $T_{1/2}$, γ-ray branching ratios, multipolarites.

^{35}Cl Levels

E(level)†	J$^\pi$‡	$T_{1/2}$	Comments
0	3/2$^+$		
1763.19 *10*	5/2$^+$		
2645.45 *12*	7/2$^+$		
3001.98 *22*	5/2$^+$		
3162.64 *9*	7/2$^-$	29.0 ps *13*	$T_{1/2}$: from τ=41.8 ps *18* (1976Wa11).
3942.3 *10*	9/2$^+$		
4347.43 *17*	9/2$^-$		
5406.68 *19*	11/2$^-$		
5926.47 *29*	11/2$^-$		
6086.89 *23*	13/2$^-$	5.3 ps *10*	$T_{1/2}$: from τ=7.7 ps *14* (1976Wa11).
7872.78 *30*	13/2$^+$		$T_{1/2}$: 0.5 ps<τ<50 ps (1976Wa11).
8844.2 *4*	17/2$^+$	5.5 ps *14*	$T_{1/2}$: from τ=8 ps *2* (1976Wa11).

† From a least-squares fit to γ-ray energies.
‡ As given in 1976Wa11 based on measured $\gamma(\theta)$ and known assignments of low-lying levels. When considered in Adopted Levels, firm assignments here will be placed in parentheses if there are no strong supporting arguments.

γ(^{35}Cl)

A_2 and A_4 under comments are from $\gamma(\theta)$ in 1976Wa11, with A_4=0 if not given.

E$_\gamma$†	I$_\gamma$†	E$_i$(level)	J$^\pi_i$	E$_f$	J$^\pi_f$	Comments
160.66# *20*	2.3	3162.64	7/2$^-$	3001.98	5/2$^+$	A_2=−0.24 *15*.
517.26 *10*	<2.2‡	3162.64	7/2$^-$	2645.45	7/2$^+$	E$_\gamma$,I$_\gamma$: doublet with γ rays in ^{36}Cl (1976Wa11). A_2=+0.47 *25*.
680.22 *15*	25	6086.89	13/2$^-$	5406.68	11/2$^-$	A_2=−0.20 *2*.
882.32 *35*	2.3	2645.45	7/2$^+$	1763.19	5/2$^+$	A_2=+0.31 *40*.
971.38 *20*	8.0	8844.2	17/2$^+$	7872.78	13/2$^+$	A_2=+0.28 *7*, A_4=−0.18 *7*.
1059.17 *20*	4.6	5406.68	11/2$^-$	4347.43	9/2$^-$	A_2=−0.04 *15*. %Branching=13 *4*.
1184.80 *20*	21	4347.43	9/2$^-$	3162.64	7/2$^-$	A_2=−0.66 *4*, A_4=+0.08 *4*. %Branching=69 *5*.
1579.15 *30*	1.9‡	5926.47	11/2$^-$	4347.43	9/2$^-$	A_2=−0.6 *4*.
1701.85 *25*	9.5	4347.43	9/2$^-$	2645.45	7/2$^+$	A_2=−0.20 *3*. %Branching=31 *5*.
1739.3 *4*	1.7	6086.89	13/2$^-$	4347.43	9/2$^-$	
1763.15 *10*	32	1763.19	5/2$^+$	0	3/2$^+$	A_2=−0.28 *2*, A_4=+0.05 *3*.
1786#	0.63	7872.78	13/2$^+$	6086.89	13/2$^-$	E$_\gamma$,I$_\gamma$: inferred from the presence of 680γ-971γ coin (1976Wa11). %Branching=8 *5*.
1946.40 *30*	1.9	7872.78	13/2$^+$	5926.47	11/2$^-$	A_2=−0.2 *1*. %Branching=22 *5*.
2179	5.0	3942.3	9/2$^+$	1763.19	5/2$^+$	E$_\gamma$,I$_\gamma$: observed in $\gamma\gamma$-coin only (1976Wa11).
2244.00 *25*	32	5406.68	11/2$^-$	3162.64	7/2$^-$	A_2=+0.23 *2*, A_4=−0.08 *2*. %Branching=87 *4*.
2465.85 *30*	5.9	7872.78	13/2$^+$	5406.68	11/2$^-$	A_2=−0.21 *6*. %Branching=70 *5*.

Continued on next page (footnotes at end of table)

^{27}Al(^{14}N,αpnγ) 1976Wa11 (continued)

γ(^{35}Cl) (continued)

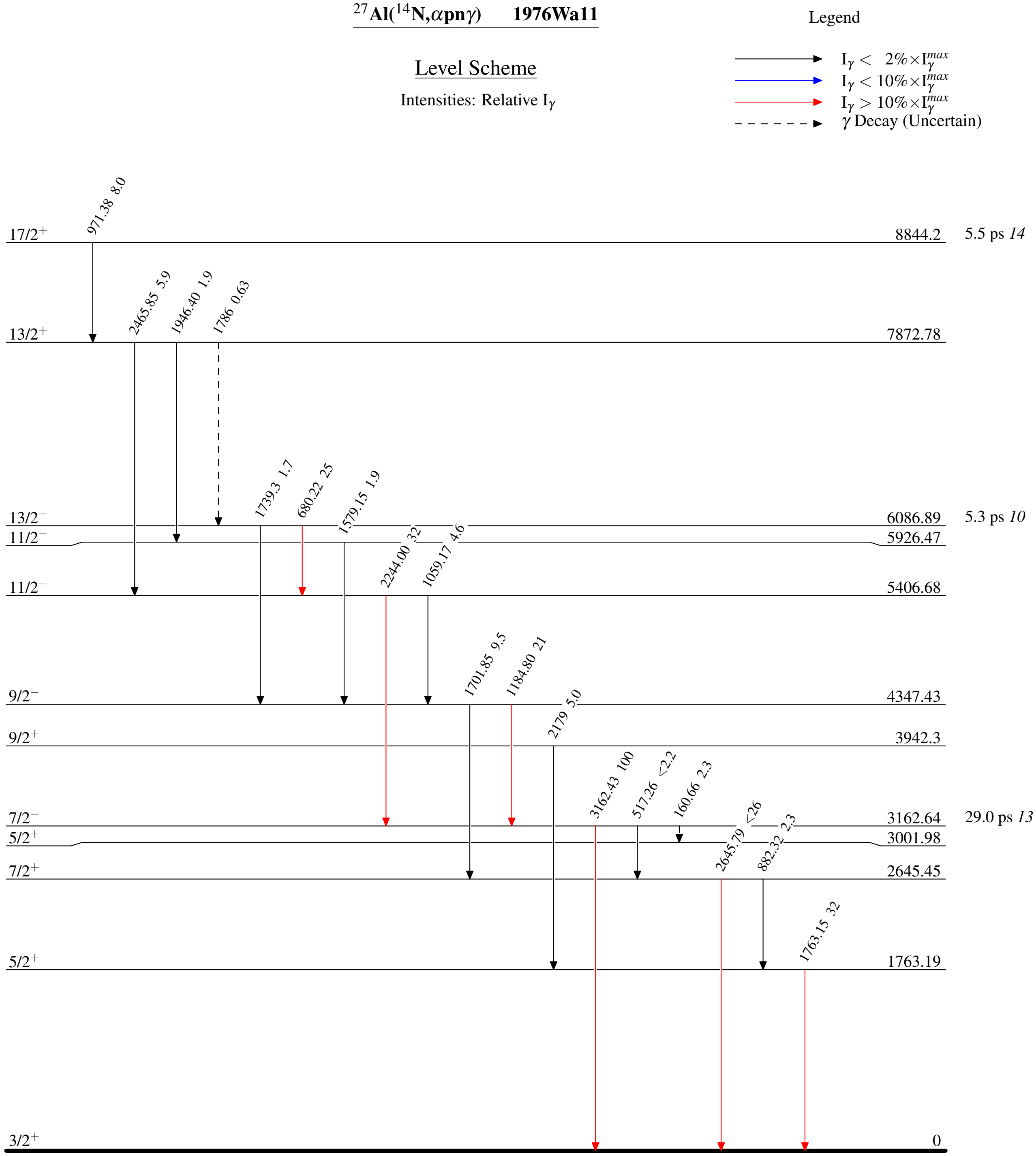
E_γ †	I_γ †	E_i(level)	J^π_i	E_f	J^π_f	Comments
2645.79 *30*	<26	2645.45	7/2$^+$	0	3/2$^+$	E_γ,I_γ: doublet with γ rays in ^{38}K (1976Wa11).
3162.43 *10*	100	3162.64	7/2$^-$	0	3/2$^+$	A_2=+0.37 *3*, A_4=−0.04 *3*.

† From 1976Wa11. Original values of relative intensities have been renormalized by the evaluator to I(3162γ)=100.
‡ Estimated from $\gamma\gamma$-coin data or known branching ratios (1976Wa11).
# Placement of transition in the level scheme is uncertain.

^{27}Al(^{14}N,αpnγ) 1976Wa11

Level Scheme

Intensities: Relative I_γ



$^{35}_{17}Cl_{18}$